

Problem 1 - Simulation of Sources from Monopoles

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Figure 1 shows the spherical spreading from a monopole. The frequency is 1 kHz. The sound pressure level has been calculated at two distances: 1 m and 4 m.

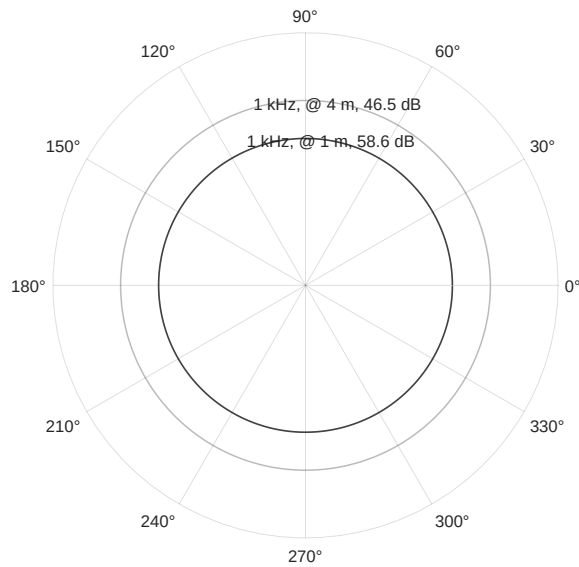


Figure 1: Monopole spherical spreading for 1 kHz. The radius of the polar plot is the sound pressure level.

Figure 2 shows the sound pressure level as a function of distance for a 1 kHz monopole source.

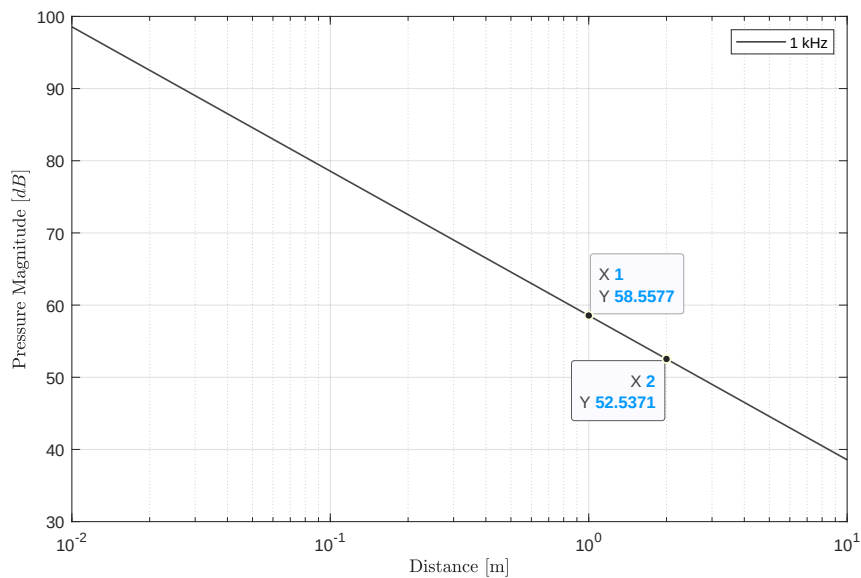


Figure 2: Monopole pressure drop versus distance for 1 kHz.

Doubling the distance yields a -6 dB of pressure level change (see data points).

Problem 1b

Figure 3 illustrates the directivity patterns for a monopole, dipole, lateral quadrupole, and a linear quadrupole for 1 kHz. The receivers were placed at a 1 m distance around each of the sources.

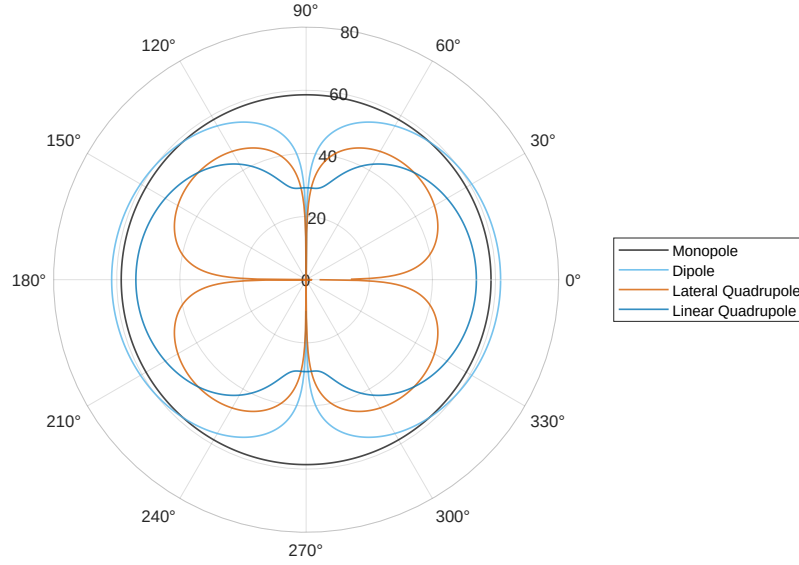


Figure 3: Directivity patterns for a 1 kHz monopole, dipole, lateral quadrupole, and linear quadrupole.

Table 1 lists the sound pressure levels for the 4 types of patterns for 1 kHz.

Source Type	Sound Pressure Level [dB]
Monopole	58.6
Dipole	61.6
Lateral Quadrupole	48.2
Linear Quadrupole	53.9

Table 1: Sound pressure levels for 1 kHz at 1 m distance.

The monopole had a volume velocity of $1.4 \frac{\text{m}^3}{\text{s}}$.

The dipole had two equal strength, in-phase elements with a separation of 0.0429 m, which is less than the 1 kHz wavelength of 0.343 m. The volume velocity for both sources was $1.4 \frac{\text{m}^3}{\text{s}}$.

The lateral quadrupole had four equal strength, diagonally opposite in-phase pairs with a separation of 0.0429 m. The volume velocity for both sources was $1.4 \frac{\text{m}^3}{\text{s}}$.

The linear quadrupole had three sources, the end sources were equal strength and in-phase, while the center source was twice the strength of the single sources and 180° out-of-phase. The separation between all three sources was 0.0429 m. The volume velocity of the end sources was $1.4 \frac{\text{m}^3}{\text{s}}$ and the volume velocity for the center source was $2.8 \frac{\text{m}^3}{\text{s}}$.

Problem 1c

Figure 4 shows a set of directivity patterns for a 9-element line source with a separated of 0.0429 m centered at the origin along the x-axis (demarcated by the 180° to 0° line in the polar plot) for 100 Hz, 2 kHz, and 5 kHz. The receivers are placed at a 1 m distance around the line source.

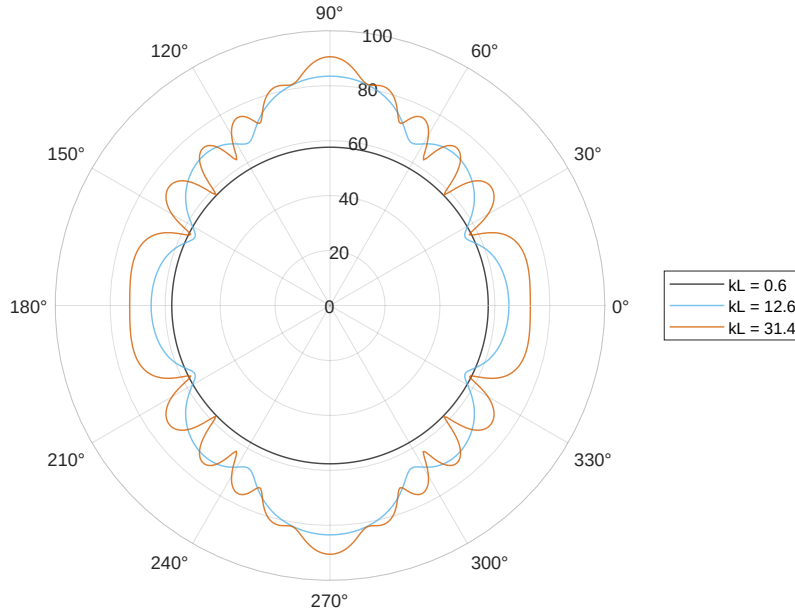


Figure 4: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with small source separation.

The length of the array is 0.343 m, or the 1 kHz wavelength. As the frequency of the source increases (i.e., $k \cdot L$), the patterns have larger directivity but with more side lobes.

Figure 5 shows the directivity patterns for a similar line source, but with the sources now separated by 0.172 m.

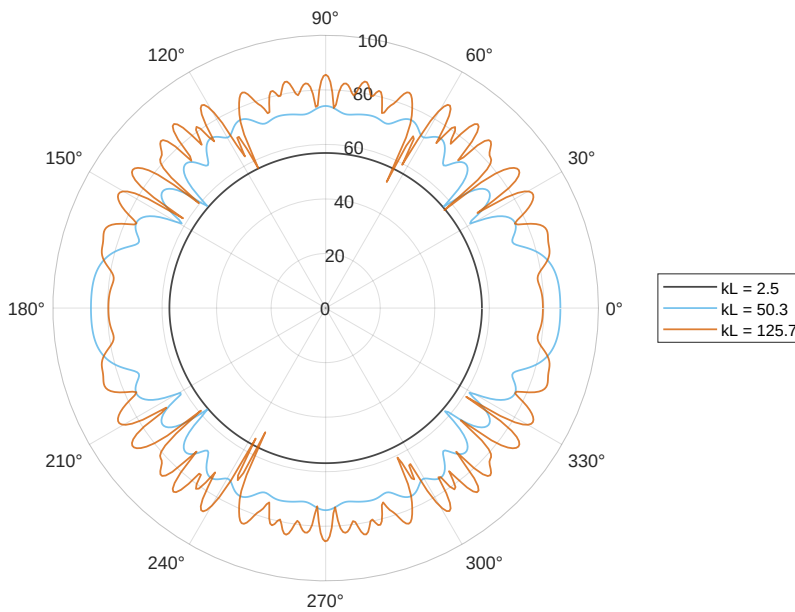


Figure 5: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with large source separation.

The length of the array is now 1.37 m, or 4 times the 1 kHz wavelength. As the frequency of the source increases (i.e., $k \cdot L$), the patterns have larger directivity but with more side lobes. Here, unlike the previous case, there are a greater number of side lobes for the 2 kHz signal (blue curve).

Problem 1d

Figure 6 shows the monopole source locations in the xy-plane. The number of sources in the plane was 100 and “a” was set to 1.715 m (5 times the 1 kHz wavelength of 0.343 m).

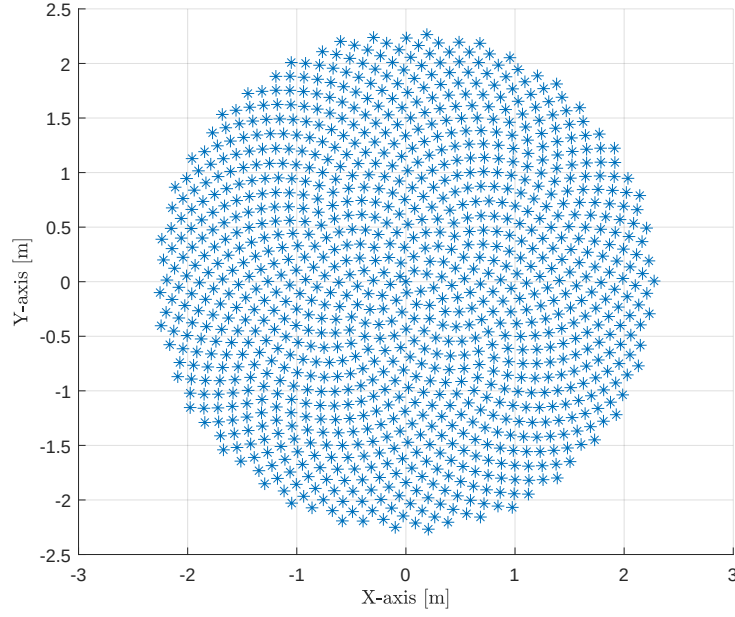


Figure 6: Monopole source locations in the xy-plane.

Figure 7 shows the estimated and theoretical null locations along the z-axis above the xy-plane for the baffled piston.

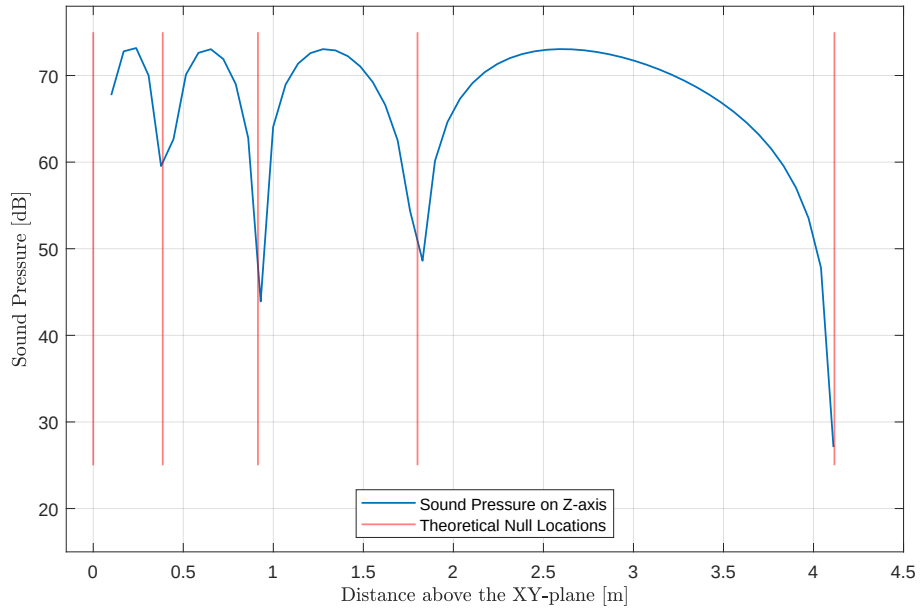


Figure 7: The estimated and theoretical null locations along the z-axis for the baffled piston.

The factor $\frac{a}{\lambda}$ was 5; there are 5 theoretical null locations. There is good agreement between the estimated and theoretical null points. The estimated null near the surface of the piston is governed by starting receiver location on the z-axis. As this distance is made smaller, the sound pressure level grows rapidly due to the calculation method of the monopole sources.

Problem 2 - Active Control of Enclosure Opening

The Matlab code for this problem is listed in Appendix ??.

Problem 2a

Figure 1 shows the sound pressure level reduction versus octave band frequency for the linear quadrupole.

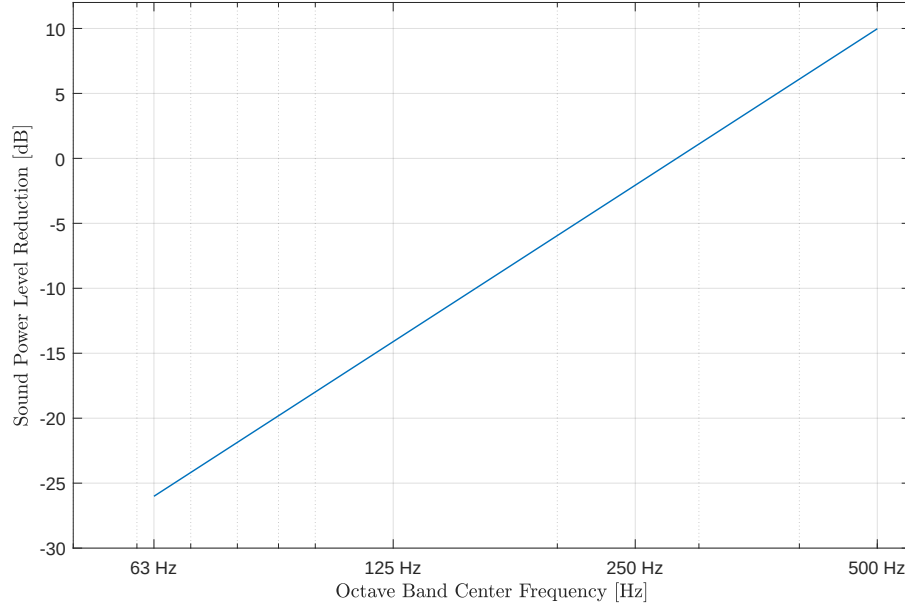


Figure 1: Sound pressure level reduction as a function of octave band center frequency.

Problem 2b

Figure 2 shows the octave sound level reduction versus the angle, θ , between the plane of the opening and a line at the center of the box opening.

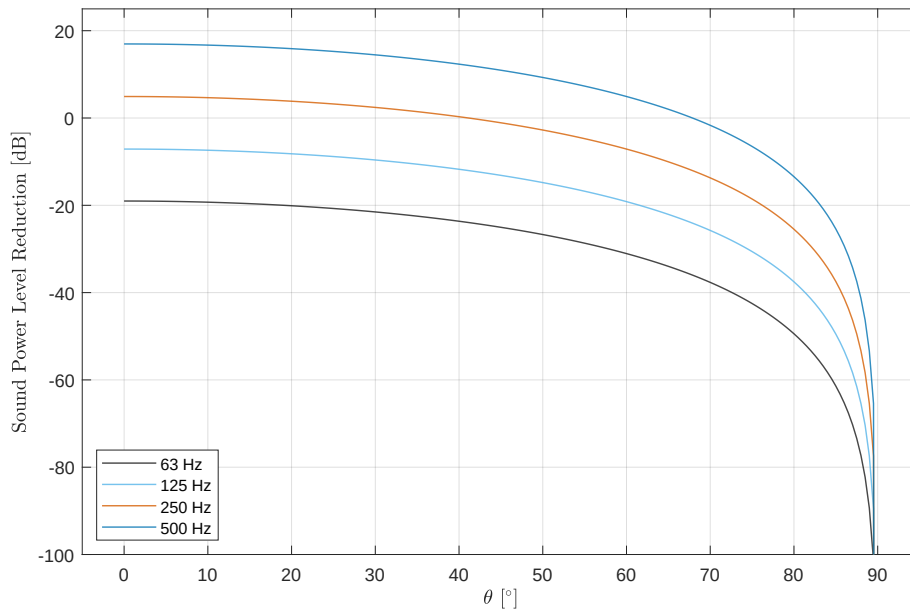


Figure 2: Octave band pressure level reduction as a function theta.

The relative level reduction across the four octave bands is similar. As the receiver moves from the side of the opening towards its center, the level of reduction increases, where at 90° , a null is reached as shown in Figure 3 for Problem 1b.

Problem 3 - Coherent and Incoherent Speakers

The Matlab code for this problem is listed in Appendix ??.

Part 3a

The expected sound pressure at the receive for the coherent sources is -17.3 dB.

Part 3b

The expected sound pressure at the receive for the incoherent sources is -0.71 dB.

Problem 4 - Traffic Noise Modeling

The Matlab code for this problem is listed in Appendix ??.

Part 4a

If the line of traffic is infinitely long, the estimated sound pressure level at the house is approximately 100 dB.

From slide 27 of Lecture 22 (Monday, April 2, 2025), the first knee-point is $\frac{D}{3}$ is 2.67 m, while the second knee-point is infinite because of the assumption of an infinitely long line of traffic. This requires that the decay per doubling of distance is -3 dB.

Figure 1 shows the estimated sound pressure level versus number of line sources.

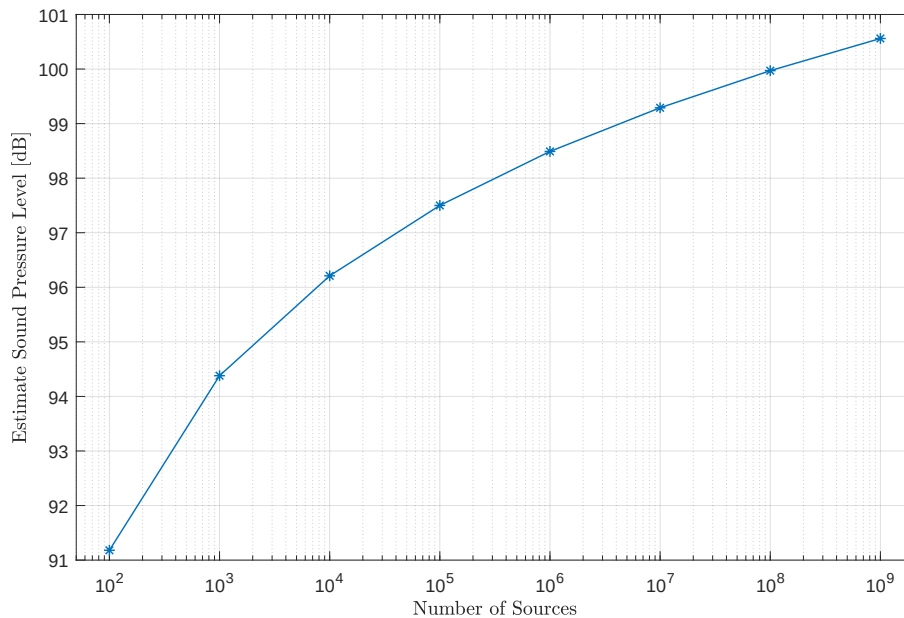


Figure 1: Estimated sound pressure level versus number of line sources.

Part 4b

When the line source circles the house at a 200 m radius, there will be a finite number of incoherent sources around the house. The number of estimate sources is 157, which produces a pressure level at the house of about 93 dB.

1 Appendix - Matlab Code for Problem 1

```
1
2
3
4 %% Synopsis
5
6 % Problem 5 — Order Tracking of Mobius Propeller Data
7
8 % See Lecture 26
9 % "D:\15 Downloads\00 GitHub\ACS_547\35 Lectures\26 Monday, April 21, 2025\Lecture 26 — Shafting and
   bearings.pptx""
10
11
12
13 %% Note(s)
14
15
16
17 %% Environment
18
19 close all; clear; clc;
20 % restore default path;
21
22 addpath( genpath( './00 Support' ), '-begin' );
23
24 % set( groot, 'DefaultFigurePosition', [ 1.7e3 775 750 500 ] ); % x, y, width, height
25
26 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
27 set( 0, 'DefaultFigureWindowStyle', 'norma' );
28 set( 0, 'DefaultLineLineWidth', 1.0 );
29 set( 0, 'DefaultTextInterpreter', 'Latex' );
30
31 format ShortG;
32
33 pause( 1 );
34
35
36
37 %% Constants and Parameters
38
39
40 color_map = slanCM( 'ColorBlind', 4 );
41
42
43
44 %% Load Data
45
46 load( 'mobius_prop_data.mat' ); % Variable(s): fs; mic_angles_degrees; p; trigger
47
48
49
50 %% Visualize Data
51
52 % for index = 1:1:11
53 % figure( ); plot( p( :, index ) ); grid on;
54 % end
55
56
57
58 %% Listen
59
60 % soundsc( p( :, 1 ), fs ); % 23 degrees
61
62 % i = 5; figure( ); plot( p( :, i ) ); grid on; soundsc( p( :, i ), fs ); % 0 degrees
63
64 % soundsc( p( :, 11 ), fs ); % -32 degrees
65
66
67
68 signal = p(:, 1);
69 % signal = p(:, 5);
70
71 frame_length = 2048; frame_hop = 1024;
72
73 APPLY_HANN_WINDOW = 1;
74 spectrogram_mrw = spectrogram_September_26_2023( signal, frame_length, frame_hop, fs,
   APPLY_HANN_WINDOW );
75
76 figure( ); ...
77 h_1 = imagesc( spectrogram_mrw.time_indices, spectrogram_mrw.Sxx.frequencies, 10*log10(
   spectrogram_mrw.Sxx.spectrum ), [ -100 -10 ] );
78 labelColorbar( 'PSD [dB re: 1 $\frac{\text{Volts}^2}{\text{Hz}}$]' ); grid on;
79 colormap parula; % Option: Turbo
80 set( gca, 'GridColor', 'w', 'GridAlpha', 0.3 );
81 title( 'Order Tracking', 'Interpreter', 'Latex' );
82 xlabel( 'Time [seconds]', 'Interpreter', 'Latex' ); ylabel( 'Frequency [Hz]', 'Interpreter', '
   Latex' );
83 set( gca, 'ydir', 'normal' );
84 ylim( [ 0 fs/2 ] );
85 % datatip( h_1, 2, +128 );
```

```

86     % datatip( h_1, 2, -128, 'Location', 'southeast' );
87
88
89
90 %% Clean-up
91
92 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
93     monitors = get( 0, 'MonitorPositions' );
94     if ( size( monitors, 1 ) == 1 )
95         autoArrangeFigures( 3, 4, 1 );
96     elseif ( 1 < size( monitors, 1 ) )
97         autoArrangeFigures( 2, 2, 2 );
98     end
99 end
100
101 fprintf( 1, '\n\n*** Processing Complete ***\n\n' );
102
103
104
105 %% Reference(s)
106
107 % p = 10 + 1i*5; % Pascals; sinusoid
108 %
109 % p_mag = abs( p ); % 11.18 Pascals
110 %
111 % p_rms = p_mag / sqrt(2); % 7.91 Pascals RMS
112 %
113 % p_dB_SPL = 20*log10( p_rms / 20e-6 ); % 111.94 dB SPL Z
114 %
115 %
116 % p_dB_SPL_verify = convert_complex_pressure_to_dB_SPL( p );
117
118
119
120
121 %% URLs

```